

$$\tan^2 \alpha + 1 = \sec^2 \alpha$$

$$\cos^2 \alpha + 1 = \sec^2 \alpha$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$$

$$\tan x = \frac{2 \tan \frac{x}{2}}{1 - \tan^2 \frac{x}{2}}$$

$$\sin 3x = 3 \sin x - 4 \sin^3 x$$

$$4 \cos^3 x - 3 \cos x$$

$$f(a) = f(b) \Leftrightarrow a = b \rightarrow \text{funktionswert zu } f \text{ ist } \text{unic}$$

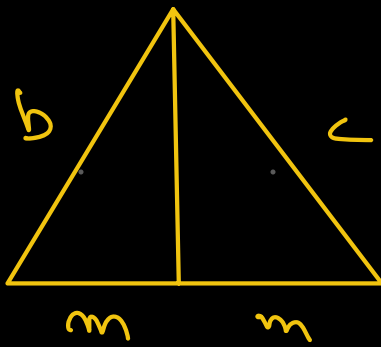
Sobry

$$\text{Im}(f) = \text{D}(f)$$

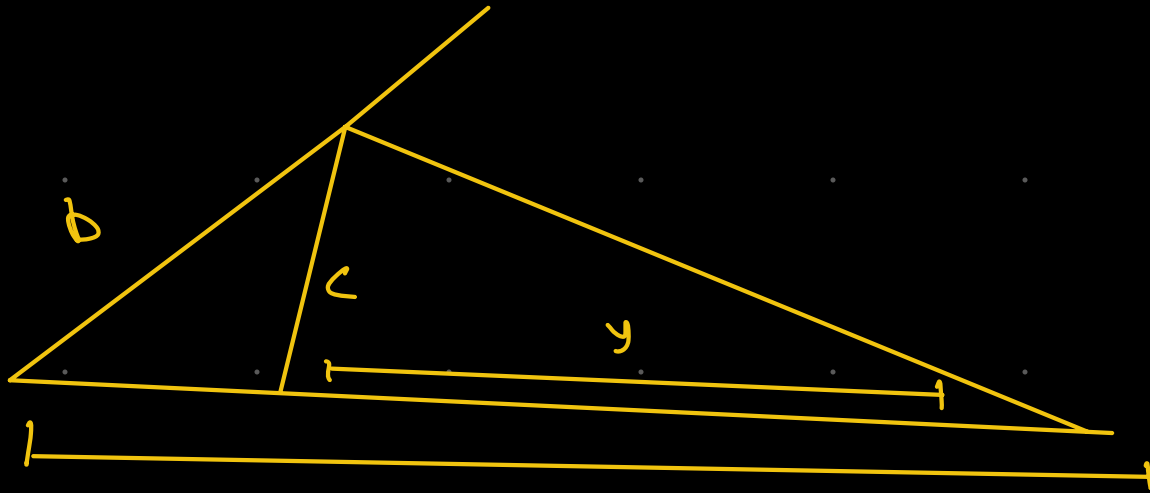
$$f(x) = 2x \quad x = \frac{y}{2}$$

Quais são todos os valores de $f(x)$ para
 assumir $\text{Im}(f)$

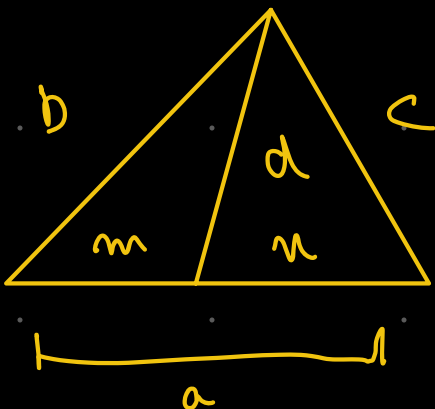
Teorema de BTZ Inter.



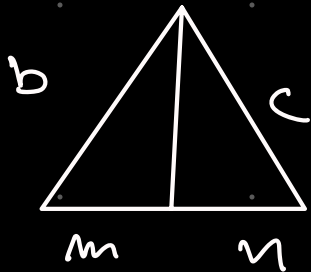
$$\frac{b}{c} = \frac{m}{m}$$



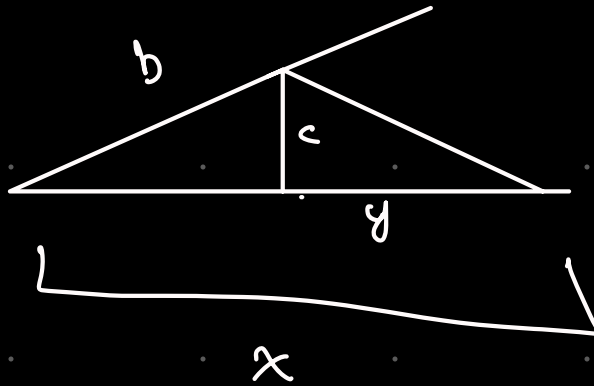
$$\frac{b}{c} = \frac{x}{y} \quad \text{Teorema de BTZ Ext.}$$



$$man + dad = bmb \quad cnc$$

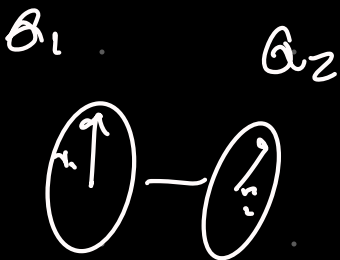


$$\frac{b}{c} = \frac{m}{n}$$



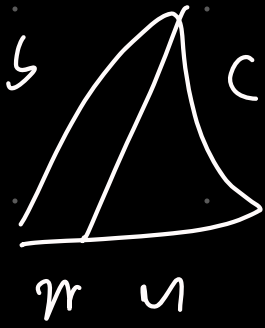
$$\frac{b}{c} = \frac{x}{y}$$

$$man + dad = bmb + cnc$$



$$V_1 = \frac{kq_1}{r_1} \quad V_2 = \frac{kq_2}{r_2}$$

$$V_{eq} = \frac{kq_1}{r_1} = \frac{kq_2}{r_2} = V_1' = V_2'$$

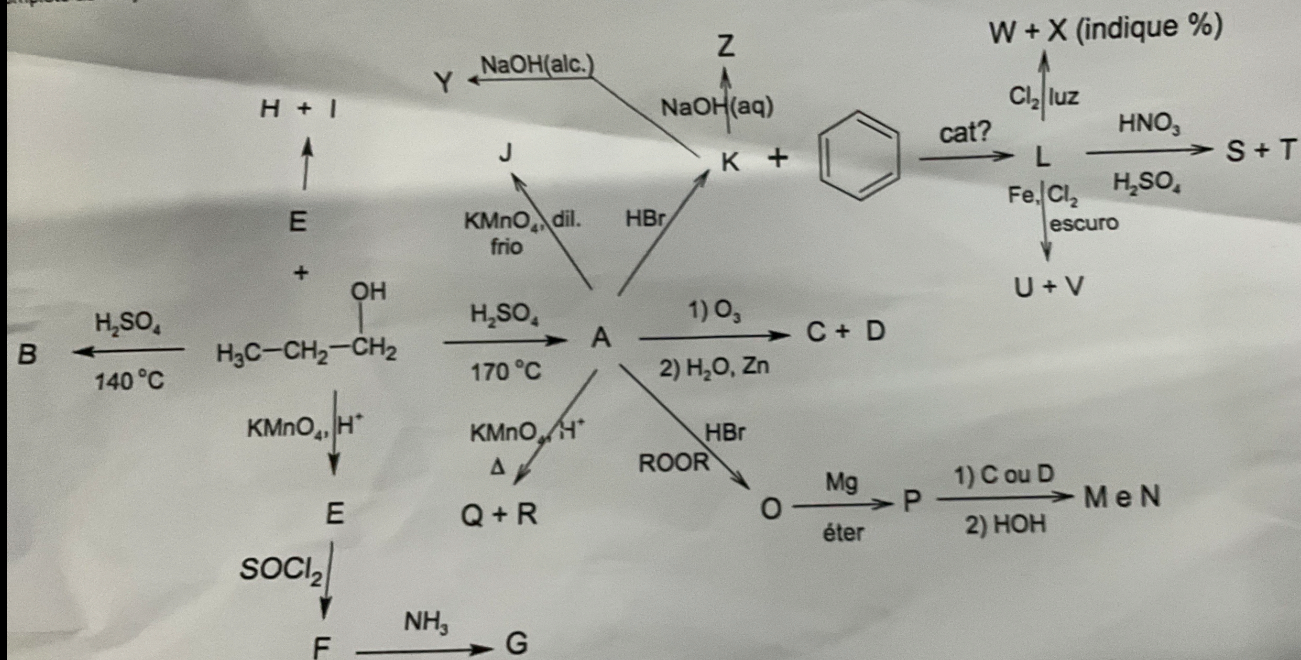


$$\frac{b}{c} = \frac{m}{n}$$

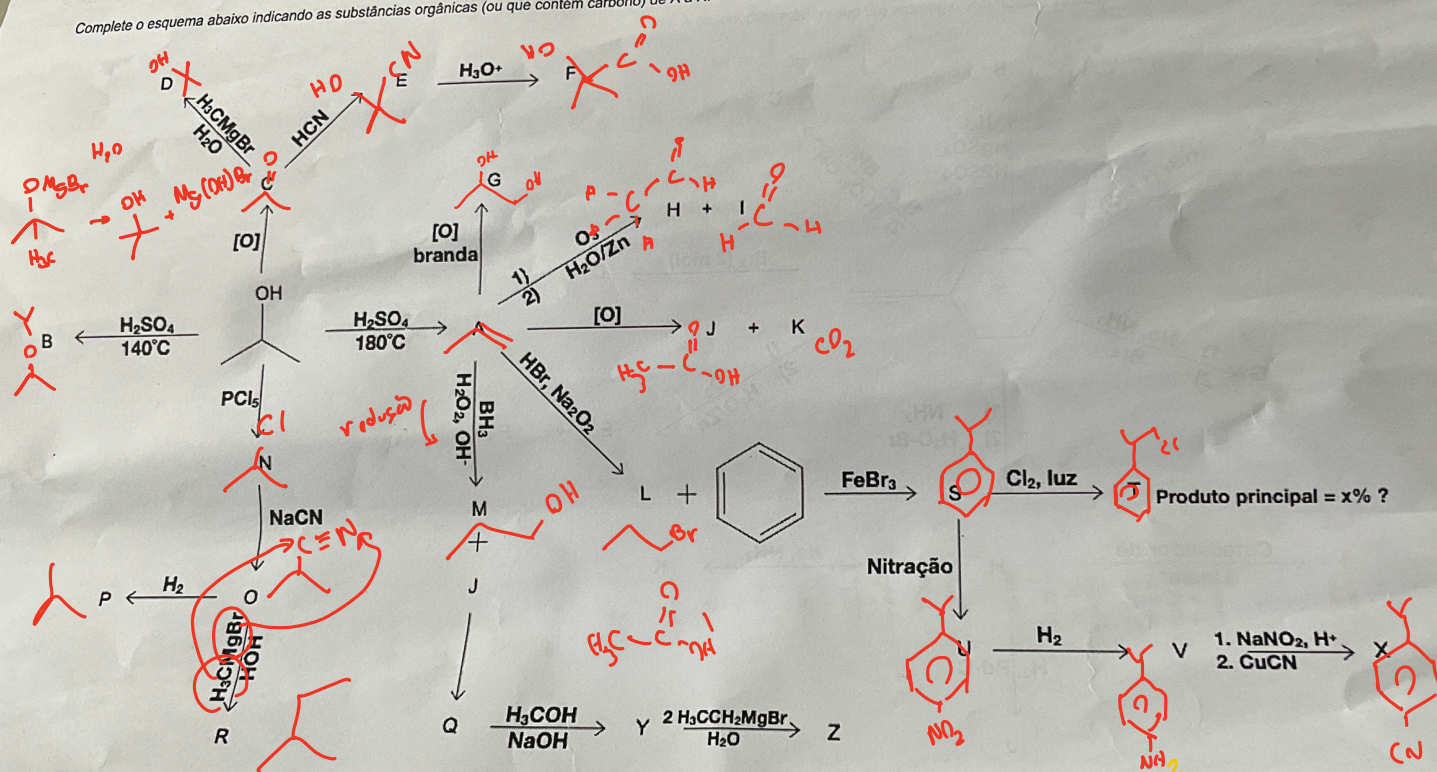


$$\frac{b}{c} = \frac{x}{n}$$

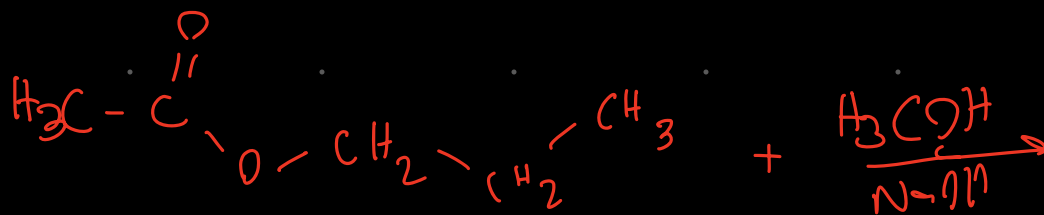
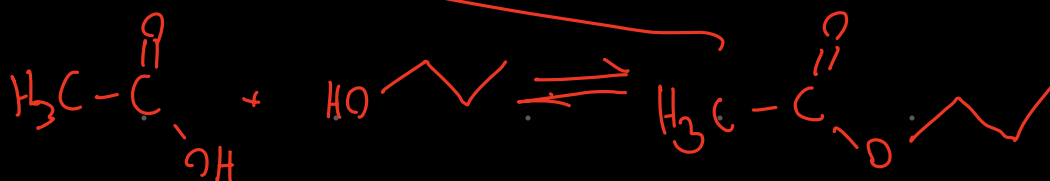
complete as reações e nomeie os compostos de A a Z.



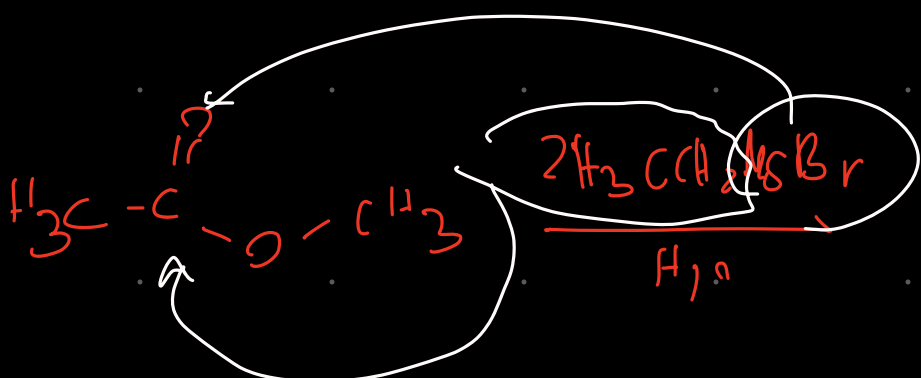
Complete o esquema abaixo indicando as substâncias orgânicas (ou que contém carbono) de A a X.

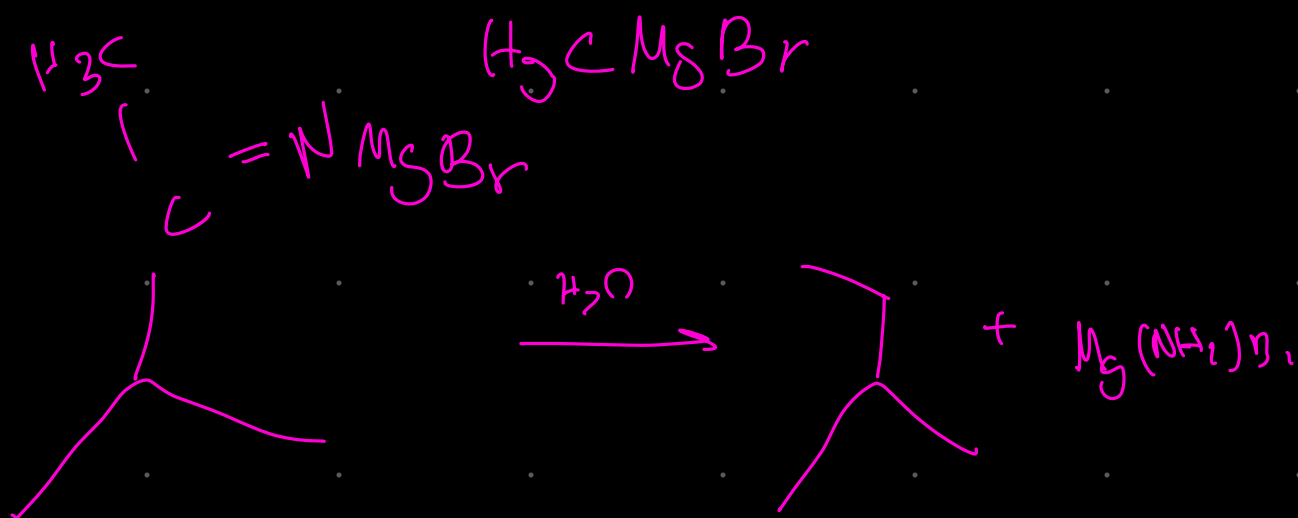
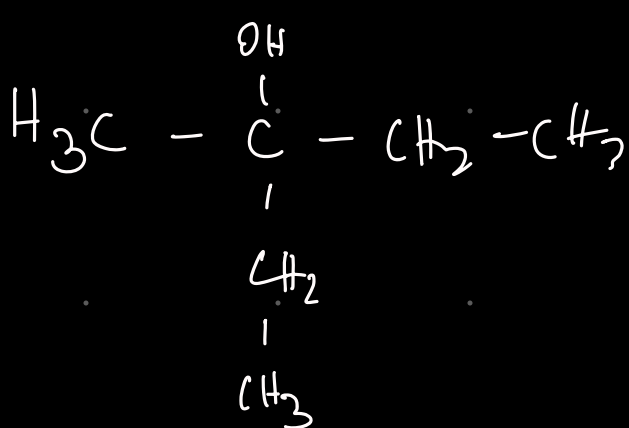


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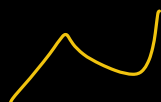
transesterificação

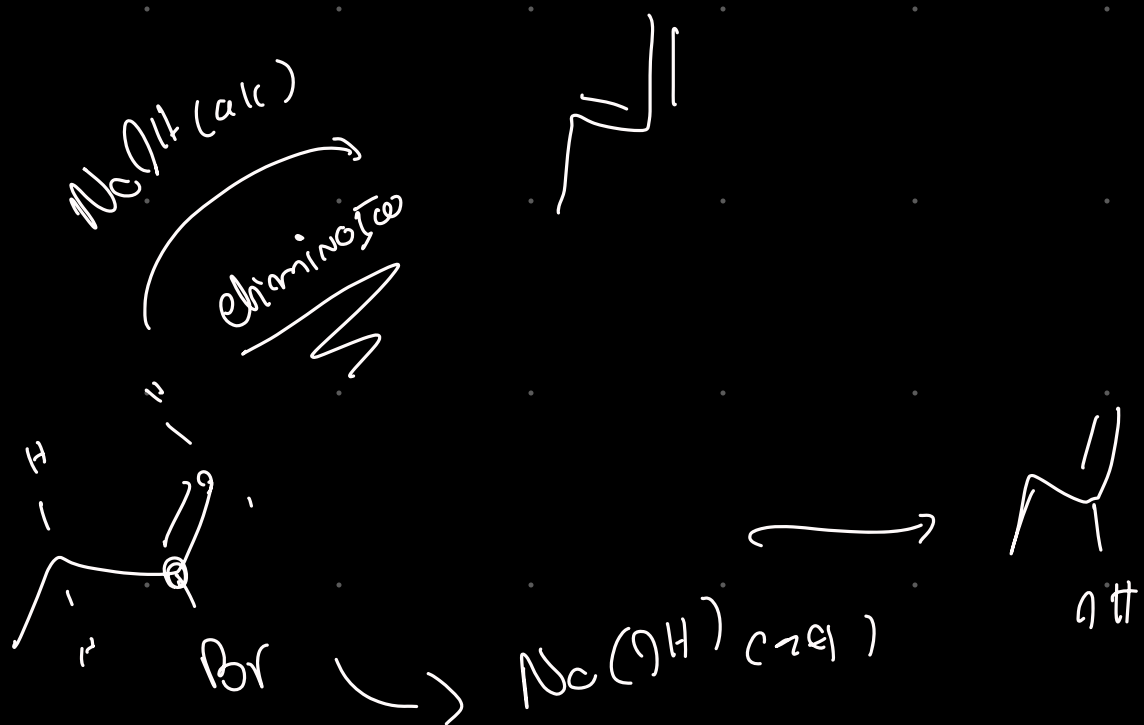


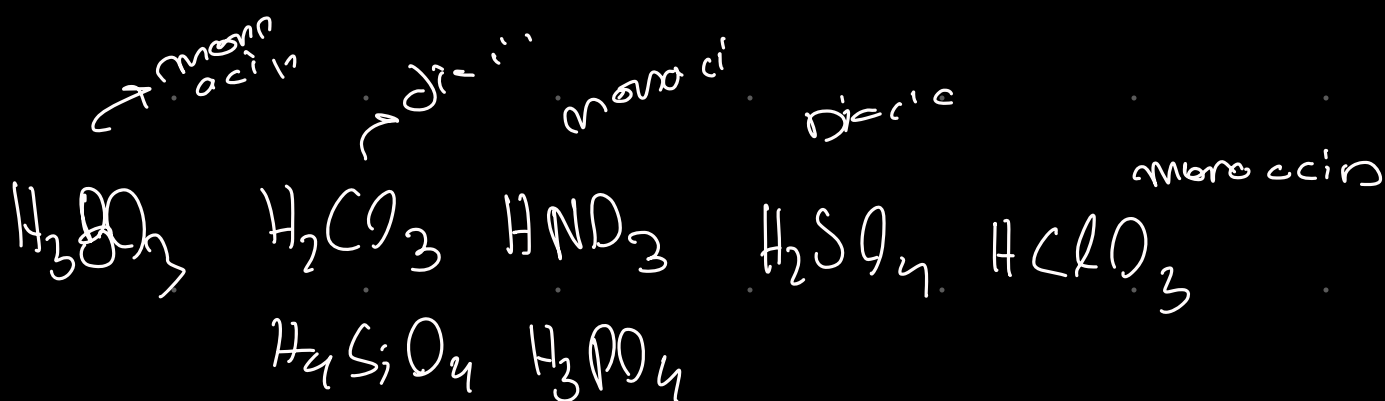
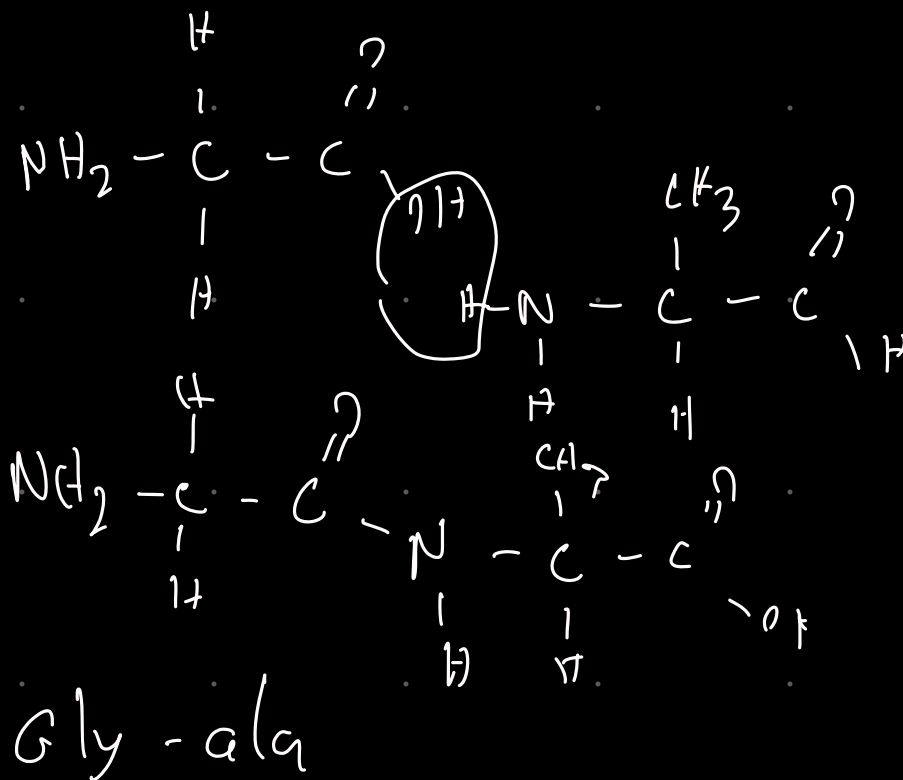


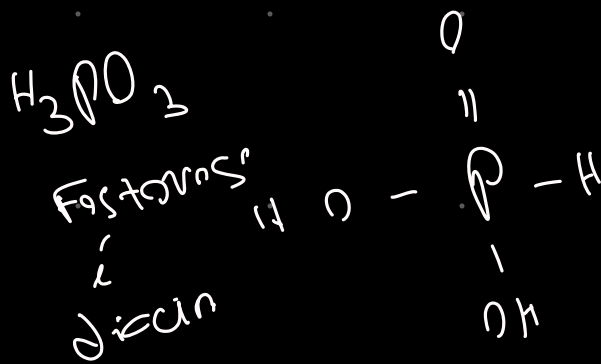
Markovnikov dá o H no qum tem mais

Zaitsev tira do qum tem menos









Oleo é Insaturado
 Gordura é Saturado

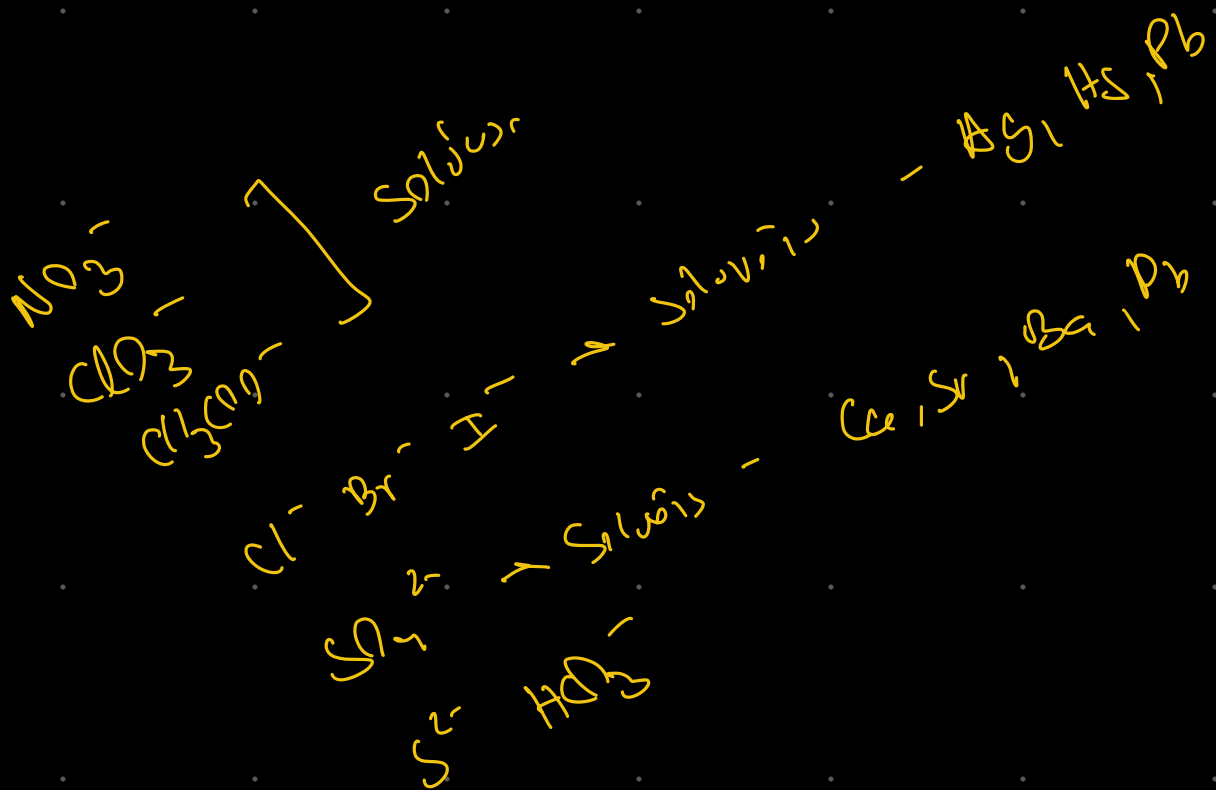
$$PC_n = (n-1)!$$

$$AC_{n,p} = n^p_{n-1+p}$$

$$CC_{n,p} = P_{n-1+p}$$

$$PK = n! \sum_{k=1}^n \frac{(-1)^k}{k!}$$

$$f(n, p) = C_{n-p+1, p}$$



$$\sin x = \frac{2Tg^{\frac{x}{2}}}{1 + Tg^2 \frac{x}{2}}$$

$$\cos x = \frac{1 - Tg^2 \frac{x}{2}}{1 + Tg^2 \frac{x}{2}}$$

$$\frac{p}{p-v} = \frac{p}{p-v}$$

$$P(z) = z^3 + az^2 + bz + c \quad \text{raizes } \omega + 3i, \omega + 9i$$

$$2\omega - 4$$

$$\omega = x + yi$$

$$\sigma_1 = -a = \omega + 3i + \omega + 9i + 2\omega - 4$$

$$\sigma_3 = -c = (\omega + 3i)(\omega + 9i)(2\omega - 4)$$

$\hookrightarrow x + yi$

$$\underbrace{x + i(y+3)} + \underbrace{x + (y+9)i} + \underbrace{2x + 2yi - 4} = \sigma_1$$

$$\downarrow \quad (4y+12)i + 4x - 4 = \sigma_1 = -a$$

$x \in \mathbb{R}$

$$a \in \mathbb{R}$$

$$4y = -12 \quad y = -3$$

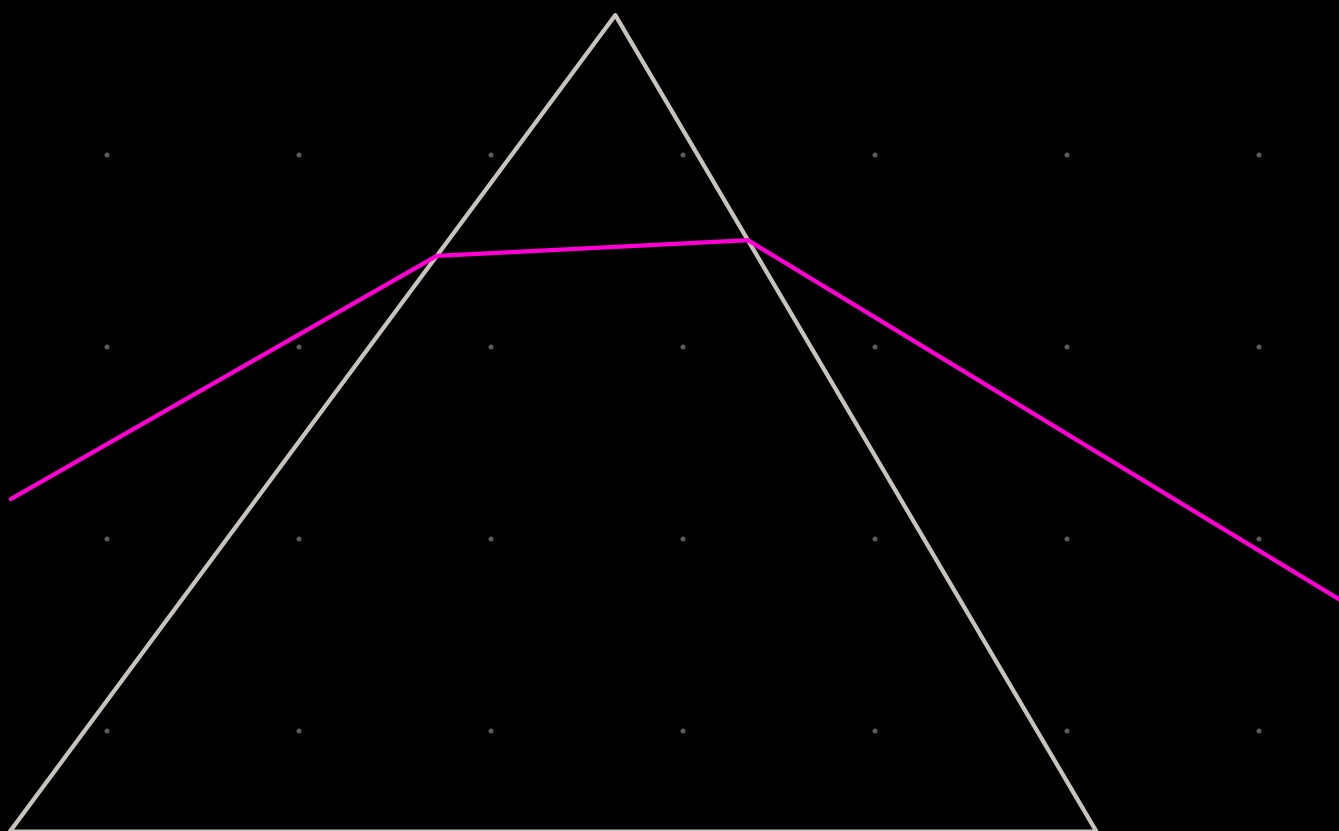
$$4x - 4 = a$$

$$2x - 4 - 6i$$

$$= x - 6i$$

$$\boxed{x = 2}$$

$$\omega = 2 - 3i$$



$$a \sin \theta = \Delta x = n_p \frac{\lambda}{2}$$

$$S_{cl}$$

$$L = \frac{m}{2} v^2$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$$

$$\frac{d}{dt} \left(\frac{\partial \dot{x}^2}{\partial \dot{x}} \right)$$

$$m \frac{d\dot{x}}{dt} = \frac{\partial \dot{x}^2}{\partial x}$$

↓

$$\frac{\partial(\dot{x}\dot{x})}{\partial x}$$

$$2\dot{x} \frac{\partial \dot{x}}{\partial x}$$

$$= m \frac{\partial \dot{x}}{\partial t}$$

$$m \ddot{x} = 0$$

$$S = \int_{t_a}^{t_b} \frac{m}{2} \dot{x} \dot{x} dt$$

$$S = \frac{m}{2} \int_{t_a}^{t_b} \dot{x} \dot{x} dt$$

$$S = \frac{m}{2} [x \dot{x}]_{t_a}^{t_b} - \int_{t_a}^{t_b} x \ddot{x} dt$$

integration by parts

Sistemas físicos possuem Energia entre

Potencial e cinética $E_c - E_p$ a diferença

entre elas varia com o tempo

além do sistema com momentos variáveis (etc).

é o sistema que observamos (classicamente)

$$(T - V) = \frac{dS}{dt}$$

onde S

é uma medida

do movimento total

ou Ação

$$dS = (T - V) dt$$

$$S = \int (T - V) dt$$

$$S = \int_{t_0}^{t_1} L(\dot{x}, x, t) dt$$

constitui a ação de um caminho

Como determinar a
menor ação

em relação ao q se
Deriva?

$$\rightarrow S[x(t)]$$

$$\frac{\delta S[x(t)]}{\delta(x(t))} = 0$$

$$\frac{\delta S}{\delta(x(t))} = \lim_{\delta x(t) \rightarrow 0} \frac{S[x(t) + \delta x(t)] - S[x(t)]}{\delta x(t)} = 0$$

um caminho pouco diferente

$$S[x(t) + \delta x(t)] - S[x(t)] = 0$$

$$\int_{t_1}^{t_2} L(\dot{x}(t) + \delta \dot{x}(t), x(t) + \delta x(t), t) dt$$

$$- \int_{t_1}^{t_2} L(\dot{x}(t), x(t), t) dt = 0$$



$$S[x(t)]$$

único
↓
 $x(t)$

$$x_L(t)$$

$$\dot{x}_L(t)$$

$$x_1(t)$$

$$\int_{t_1}^{t_2} \mathcal{L}(\dot{x} + \delta \dot{x}, x + \delta x, t) dt = \mathcal{S}[x(t)]$$

↓

$$\int_{t_1}^{t_2} \left[\mathcal{L}(\dot{x}(t), x(t), t) + \delta \dot{x} \frac{\partial \mathcal{L}}{\partial \dot{x}} + \delta x \frac{\partial \mathcal{L}}{\partial x} \right] dt$$

$$\underbrace{\int_{t_1}^{t_2} \mathcal{L}(\dot{x}, x, t) dt}_{\mathcal{S}[x(t)]} + \int_{t_1}^{t_2} \left[\delta \dot{x} \frac{\partial \mathcal{L}}{\partial \dot{x}} + \delta x \frac{\partial \mathcal{L}}{\partial x} \right] dt$$

$$\mathcal{S}[x(t)] + \underbrace{\int_{t_1}^{t_2} \left[\delta \dot{x} \frac{\partial \mathcal{L}}{\partial \dot{x}} + \delta x \frac{\partial \mathcal{L}}{\partial x} \right] dt}_A = \mathcal{S}[x(t)]$$

$$\left[\delta x \frac{\partial \mathcal{L}}{\partial \dot{x}} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta x \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} \right] dt = 0$$

$\delta x \Rightarrow$ a variação do percurso é 0 nos finais

$$\delta x \Big|_{t_1}^{t_2} = 0$$

$$\int_{t_1}^{t_2} \delta x \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} \right] dt = 0$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

euler lagrange.

Exercise 2

$$L = \frac{m}{2} (\dot{x}^2 - \omega^2 x^2)$$

$$S = \int L dt$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$$

$$2m\ddot{x} = -2m\omega^2 x$$

$$\ddot{x} = -\omega^2 \cdot x$$

$$\omega^2 = \frac{\ddot{x}}{x}$$

$$x = -kx$$

$$S = \frac{m}{2} \left[\int \dot{x}^2 dt - \int \omega^2 x^2 dt \right]$$

$$\dot{x} x \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \ddot{x} x dt - \int \ddot{x} x dt$$

$$ABC \quad BC \rightarrow ABC = 11BC + 80$$

$$100A + 10B + C = 110B + 11C + 80$$

$$100A = 100B + 10C + 80$$

$$a) 16$$

$$b) 18$$

$$c) 19$$

$$d) 20$$

$$\begin{array}{r} ABC \quad BC \\ - BC \quad 11 \\ \hline (A-B)(10-C)C \\ - BC \\ \hline (A-B)(10-C)(10) \\ \hline \end{array}$$

$$A-B=1 \quad 10-C=8 \quad C=2$$

$$A=B+1 \quad C=2$$

$$(B+1)(18)(2)$$

口カ口啡